

An In Situ Study of the Habits of Users That Affect Office Chair Design and Testing

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Objective: The purpose of this study was to perform an in situ assessment of office seating habits that influence chair testing and design.

Background: Many chair testing parameters were derived decades ago when the average weight of people within the United States was dramatically lower and the office work tasks less computer based.

Method: For the study, 51 participants were randomly selected from Brazos Valley, Texas, businesses to participate in 8-hr assessments of office seating habits. Overall results were compared with current chair testing and design assumptions. Data were collected through written survey and through data logging of seat and back contact pressure and duration with the use of the X-SENSOR™ pressure mapping device and software. Additionally, 1 day per participant of caster roll distance was recorded with the use of a caster mounted digital encoder. Participants were grouped by body mass index (BMI) and weight (BMI <35 and weight <102 kg or BMI >35 and weight >102 kg).

Results: It was determined that a significant difference did exist between the groups in mean seat time per shift ($p < .001$), back cycles per shift ($p < .002$), seat cycles per shift ($p < .01$), and caster distance rolled per shift ($p < .001$).

Conclusion: Several key parameters and assumptions of current chair test methods and design specifications may no longer be valid for the upper quartile of weight range of the current U.S. population.

Application: The data collected in this study will enable engineers to determine whether revision of design standards for testing office seating for both normal weight and extremely obese workers is indicated.

Keywords: biomechanics, anthropometry, work physiology, office seating, body mass index, chair design, chair testing

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INTRODUCTION

The current test standards of office seating in the areas of quality, durability, safety, and performance in the United States are provided by the American National Standards Institute/Business and Institutional Furniture Manufacturers Association (ANSI/BIFMA). Many of the field-use assumptions that were established in offices before the proliferation of the personal computer and the dramatic increases in individual body weight witnessed in the past 30 years are still built into the current standard (ANSI/BIFMA, 2003). The industry is currently exploring two new potential standards (General Purpose–Heavy Duty Office Seating GPHD X5.11-201X and Bariatric Seating Units BAS8.1-201X) that would expand the coverage of the current tests to even larger individuals and potentially longer periods of use. The purpose for the collection of these data was to begin to create a basis for developing updated standards that reflect this change in the population.

The combined percentage of overweight or obese in the U.S. adult population has soared to more than 67% in the past few years (National Health and Nutrition Examination Survey [NHANES], 2005-2006). The most common way of measuring changes in obesity prevalence over time, for large populations, is by the body mass index (BMI; World Health Organization [WHO], 2002). A person's BMI is determined by dividing weight in kilograms by height in meters squared. Clinical guidelines from the National Institutes of Health (NIH) define overweight as a BMI of 25 kg/m² to 29.9 kg/m²; obesity is defined as a BMI of 30 kg/m² or more, and severely obese is defined as a BMI of more than 35 kg/m² (NIH, 1998). Table 1 shows the resultant shift in mean weights by gender and BMI categories from 1960 to 2004 (NHANES, 2005-2006).

TABLE 1: Changes for U.S. Adults 1960–2004 in Mean Weight by Gender and Body Mass Index (BMI)

Variable	1960	2004	Change
Average male weight	75.6 kg	86.6 kg	↑14%
Average female weight	63.7 kg	74.3 kg	↑17%
BMI >25	31.5%	32.2%	↑0.7%
BMI >30	13.4%	35.1%	↑21.7%
BMI >40	0.9%	6.2%	↑5.3%

Office Seating

Seating, as we know it, is a modern convenience. Much of the world today and, in fact, throughout much of recorded time, had very little to do with sitting during work. Some have even suggested that chairs may be at the root of our ergonomic injury or illness and obesity trends because we typically approach the problems of seated work assuming the worker should be seated in a chair in the first place (Gurr, Straker, & Moore, 1998). Today, seating is designed for institutional purposes, for the home, for general consumer use, and even as a form of artistic expression.

Functional and Life Cycle Testing of Seating

The modern office chair is typically designed and tested to pass the ANSI/BIFMA X5.1-2002 Guidelines for Testing of Chairs. This guideline is set up to define functional and proof tests of seating in both static and dynamic modes. The dynamic tests are based on forces derived from office worker cycle counts and durations of an 8-hr shift, for 40 hr per week and 10 total years of use. The 20-section test document specifies how much force should be applied, in what location, and for how much time or how many cycles. Some of the tests are functional testing (no ill effect) or proof testing (no catastrophic failure, i.e., bend but do not break) as criteria for pass or fail. It is our contention that many key points in the test parameters in Sections 5, 8 through 10, 13, 14, 17, and 20 were determined from best practices that predate the modern use of computers in office work and obese worker trends. These tests involve the use of a 102-kg load (or, for the arms, a derivative of that simulated

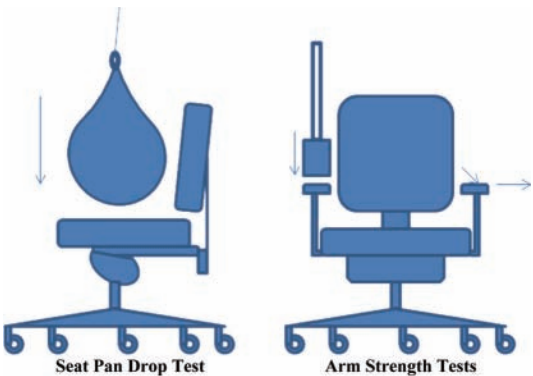


Figure 1. American National Standards Institute/Business and Institutional Furniture Manufacturers Association X5.1-2002 chair testing examples (details in Table 2). Seat pan drop test (left) and arm strength tests (right).

body weight), which was at the time intended to capture the weight of more than 95% of the U.S. adult population. There are currently no standards or test parameters specifically for seating designed for workers who significantly exceed that weight limit.

Figure 1 provides a few examples of the type of tests used in the ANSI/BIFMA standard. In Table 2, a matrix of the current test standard is provided that includes details of the basic parameters of each test. The shaded boxes in Table 2 represent the details of current count and cycle data, which this study explored. There has been little debate among industry experts as to whether the weights and forces applied in the current test should be more reflective of the increase in mean body weight depicted in Table 1.

The industry lacks consensus on whether the heavier population of office chair users is also presenting that increased weight in significantly different duration and frequency than previously used to simulate chair life. This knowledge would enable manufacturers to more accurately simulate real-world conditions and thereby design and produce seating that is the safest possible for current users and their environments. In our opinion, additional debate about the need for a test standard specific to products intended for bariatric users is also needed.

In automotive and wheelchair seat design, tests are conducted with 100-kg, humanlike test dummies (International Organization for

TABLE 2: ANSI/BIFMA X5.1-2002 General Purpose Office Chairs, Test Summary

Section	Standard Test for 102-kg, 95th-Percentile Male	Current Test Parameters
1, 2, 3, 4	Definitions and scope	Types of office chairs, measurements, tolerances, test report format
5	Backrest strength, static, Type I	Functional and proof loads of 890 N and 1,334 N are applied with locked mech at standard rear tilt angle for 1 min each
6	Backrest strength, static, Types II and III	Functional and proof loads of 667 N and 1,112 N are applied with locked mech at locked upright position for 1 min each
7	Base test	11,120 N for 1 min, 2 cycles
8	Drop test, dynamic	102-kg shot bag dropped from 15 cm at 5 cm forward of center depth of seat for 1 cycle
9	Swivel test, cyclic	102-kg weight in CG of seat swiveled 360° for 120,000 cycles
10	Tilt mechanism test, cyclic	102-kg weight in the seat is rocked just short of front and back stops for 300,000 cycles
11	Seating durability test, cyclic	57-kg weight dropped from 2.5 cm for 100,000 cycles over seat center and then 20,000 cycles on each front corner at 734 N
12	Stability test	Forces required to tip chair forward and rearward with 79 kg in chair
13	Arm strength test, vertical, static	Functional and proof vertical loads are applied over 12.7-cm-long area on the arms for 1 min at 890 N and 1,334 N, respectively
14	Arm strength test, horizontal, static	Functional and proof horizontal loads are applied over 12-cm-long area on the arms for 1 min at 445 N and 667 N, respectively
15	Backrest durability test, cyclic, Type I	445 N pulled against rearward tilt stop for 80,000 cycles and then 20,000 to each side at 10 cm from centerline
16	Backrest durability test, cyclic, Types II and III	334 N pulled against upright tilt stop for 80,000 cycles and then 20,000 to each side at 10 cm from centerline
17	Caster/chair base durability test, cyclic	Base only is loaded with 102 kg and rolled back and forth across three bumps on a 2-m table for 2,000 cycles and then bumps are removed and base is rolled for 98,000 cycles
20	Arm durability test, cyclic	60,000 cycles at 10° out and down using 400 N

Note. ANSI/BIFMA = American National Standards Institute/Business and Institutional Furniture Manufacturers Association. Shaded boxes are specifically explored in this study. Sections 18, 19, 21, 22, and 23 were omitted for clarity, as they include specifics related to chair accessories and four-legged side chairs not covered in this study. Mech = mechanism (the mechanical connection between seat, back, and height adjustment that allows for multiple access of movement); CG = center of gravity.

Standardization [ISO], 1992). Because the automotive and wheelchair manufacturing industries have realized the same needs for accommodation as the office furniture industry, they have begun to call for augmentation of the 100-kg ISO test dummy with higher mass. Specifically, Cooper and O'Connor (1999) suggested that testing needs to occur up to 250 kg, with the greatest need being an immediate switch to testing with a 150-kg dummy.

No data were found regarding the seating habits of the office worker as they pertain to the critical

cycle and load data of the ANSI/BIFMA test guidelines. Research related to the short-term testing of seating was found in pressure mapping studies. One study examined multiple male and female users in multiple seat styles and found that seat design was the most critical parameter in determining seat interface pressure (Vos, Congleton, Moore, Amendola, & Ringer, 2006). Others found that seat shape, working conditions, individual size, and individual preferences were all acceptable predictors of user comfort, but not necessarily product performance,

over time (Coleman, Hull, & Ellitt, 1998; Hjelm et al., 2000; James, 2003).

Pressure mapping systems were validated by Stinson, Porter, and Eakin (2003a) in a study that tested 65 occupational therapy students to evaluate two different pressure maps of the same individual to test for interrater reliability of data collection and interpretation between students. A pressure mapping survey was also conducted in which 63 student volunteers (44 women and 19 men) were tested at various recline angles (10°, 20°, 30°), foot support, and foot elevation. Average pressure was found to have a significant positive correlation with BMI ($r = .381$, $\alpha = .01$ level; Stinson, Porter, & Eakin, 2003b). Pressure mapping was also shown to be a reliable method of data collection for in-chair movement and was therefore used in this study (Fenety, Putnam, & Walker, 2000; Vos et al., 2006).

Another study was conducted to determine whether BMI influenced seat interface pressure in elderly persons. In this study, peak seat interface pressure was highest in the thin group with the lowest BMI levels (Kernozek, Wilder, Amundson, & Hummer, 2002). Differences in peak seat interface pressures were progressively smaller as BMI increased. In this case, high BMI seemed to protect from the development of decubitus ulcers.

Egress for seating has been reviewed in many studies. Alexander, Koester, and Grunawalt (1996) found that lowered seat height, increased posterior seat tilt, and seat compressibility were all found to significantly affect rise difficulty in the 29 older versus 21 younger volunteers studied for ease of egress. These factors were also found to correlate positively with seated comfort in another study (Boonstra et al., 2006). Finlay, Bayles, Rosen, and Milling (1983) found that 77% of those who were unable to rise without assistance could rise unaided if the arm and seat height were both placed at 25.4 cm and 45 cm, respectively.

A biomechanical analysis of foot placement during egress found that seat pans should be of adequate height (50 cm) as well as have sufficient space under the seat to permit the backward movement of the lower legs (Kawagoe, Tajima, & Chosa, 2000). This is supported by the Sibella, Galli, Romei, Montesano, and Crivellini

(2003) study that evaluated 10 normal-BMI participants and 40 obese-BMI participants during chair egress. Using optoelectronic body coordinate recording and a force platform, they were able to show that the normal group used a trunk flexion method, whereas 100% of the obese group limited their trunk flexion. This supports our overall premise that simulating chair use for obese workers is not necessarily the same as that indicated for normal-weight workers.

Some work has been done on the quantitative prediction of body diameter in severely obese individuals that might provide the basis for simple, non-gender-specific, non-race-specific models that could be applied to chair testing. This work may be most helpful to furniture designers and/or specifiers attempting to fit or design a specific individual or group of individuals, but it also offers some insights, such as body diameter prediction based on weight, that might be of use for furniture test dummy design (Fontaine et al., 2002).

A call center study evaluated workers for basic BMI stats and found that there was a 6% shift toward obesity compared with the overall Centers for Disease Control numbers for U.S. adults (Bossen, 2005). The most significant finding of this study was the report from the author that nearly one in nine of these workers (or approximately 11% of this population) exceeded the current BIFMA test criteria of a standard chair, which is listed in the ANSI/BIFMA X5.1-2002 guideline as 102 kg.

The goal of this study was to determine whether a significant difference exists between chair-use factors for Group 1 (users covered by current design and test standards, with a BMI <35 and weight <102 kg) and Group 2 (BMI >35 and weight >102 kg) participants related to seating design and testing and whether any potential differences should be included in the planned updating of chair test and design standards for office seating.

METHOD

Participants

Following Texas A&M University Institutional Review Board approval and participant consent, 51 participants were randomly selected from a survey database of more than 200 office

TABLE 3: Participant Descriptive Data

Demographic	Group 1	Group 2	Total
Gender	16 females, 8 males	16 females, 11 males	32 females, 19 males
Mean age	37.3	36.8	37
Race	20 Caucasian, 2 Hispanic, 1 African American, 1 Asian	17 Caucasian, 5 Hispanic, 5 African American	37 Caucasian, 7 Hispanic, 6 African American, 1 Asian
Height	155–185 cm; mean 174 cm	152–188 cm; mean 175 cm	152–188 cm; mean 175 cm
Weight	47.7–95.5 kg; mean 77 kg	102–150 kg; mean 131 kg	47.7–150 kg; mean 108 kg
Job type	12 clerical, 8 management, 4 technical	17 clerical, 4 management, 6 technical	29 clerical, 12 management, 10 technical

workers in the Brazos Valley of Texas that had been formed for a field trial related to seating habits of office workers.

For this study, participants were randomly selected from the 200-person office worker group and then were screened for gender, BMI, and weight. The following information was collected:

- 1. Gender
- 2. Age
- 3. Race-ethnicity
- 4. Height (cm)
- 5. Weight (kg)
- 6. Resultant BMI (hidden from participants)
- 7. Date of survey
- 8. Office classification: clerical-support, technical, or management

BMI cutoffs were chosen to mimic the shift in obesity prevalence between current U.S. workers and those of 30 years ago when chair test standards were developed. The BMI values we selected also established a group of 24 individuals well within the weight parameter of the current test standard along with a group of 27 outside the weight parameter. References herein to Group 1 and Group 2 are per these criteria of a range and not indicative of a particular category on the BMI scale. We recognize that a very tall person with Group 1 BMI could exceed 102 kg, but our interest was in individuals that were both over the 102 kg weight limit and at or above the 35 BMI level. Participant data for this study are summarized in Table 3.

Apparatus

For the back, seat, and duration counts, a stock model NP16800 chair (Neutral Posture, Inc.,



Figure 2. X-SENSOR™ array on test chair.

Bryan, TX) was used to collect data (see Figure 2). This chair was carefully selected as representative of the Type 1 and Type 2 tilting and tilting-backrest chairs described in the ANSI/BIFMA standard, but it was also chosen for its 225-kg rating because we expected multiple participants would be at or more than 180 kg. Aside from its relative size and weight capacity, it shared the other common features found in the class of chairs covered by the standard. The chair was 66 cm wide and 51 cm deep, and the backrest was 58 cm wide and 91 cm tall. The chair

has height- and width-adjustable arms along with seat pan angle and back angle controls. The seat height remained adjustable across a 42- to 53-cm range from seat pan to floor. Participants were encouraged to make the chair comfortable after being instructed in its use.

During the seat and back tests, a pressure mapping array was sewn into seat covers and positioned with its grid at the (0,0) reference point of bottom of backrest and back of seat pan. The pressure mapping system was the X-SENSOR™ system of capacitive pressure mapping for both seat and back (X-SENSOR Technology Corporation, Calgary, Alberta, Canada; ± 2 mmHg). The X-SENSOR is equipped with software that analyzes eight metrics in 2-D and 3-D graphic readouts with complete pressure analysis coming from 1,296 capacitive sensors in each pressure pad. This device was calibrated per the manufacturer's recommendations prior to each new participant. Data were stored and collected on a laptop attached to the X-SENSOR that was stored under the rear of the seat pan.

Information was collected from a digital caster roll distance evaluator designed by the researchers using a digital distance encoder that tracks caster movement in roll frequency and distance per day. This digital counter logs travel by the rear caster of the chair, which is kept in constant orientation after replacement of the standard casters with the encoding casters. The caster records in the same manner as a distance wheel used for measuring linear distance.

The digital distance counter was validated in a chair manufacturer's test lab by comparing a known travel distance on the manufacturer's machine with device readings during the course of five 100-cycle tests (per ANSI/BFMA X5.1-2002: 152 cm per cycle, or 76 cm down and 76 cm back, with a 102.2-kg weight centered over the five-leg base). Accuracy across the five test runs was $\pm 3\%$. The purpose of this test was to determine whether modern workers had different roll distances than the assumed distances used in the current chair testing standard and whether the roll distances varied between Group 1 and Group 2 workers. Underestimates of the distance traveled by chair casters could affect the durability level (given that most

companies design to those levels) and thereby the safety of the chairs that use the casters.

Procedure

The data for this research were collected in real time, in situ, in the office of each participant during normal office hours on randomly selected days of the week (Fenety et al., 2000). Workers were randomly selected by workstation location. In the end, 24 Group 1 (BMI < 35 and weight < 102 kg) and 27 Group 2 (BMI > 35 and weight > 102 kg) participants were included. Each participant sat for at least one full shift in the custom test chair designed for this research. The authors were trained in the use of the device by representatives of X-SENSOR. The device measured eight metrics, once per second for the full 8-hr shift. The metrics are as follows:

1. Seat pressure average (mmHg)
2. Seat pressure peak (mmHg)
3. Seat contact area (cm²)
4. Back pressure average (mmHg)
5. Back pressure peak (mmHg)
6. Back contact area (cm²)
7. Elapsed time (hours, minutes, seconds)
8. Center of applied pressure (2-D on seat pan and backrest)

The data collected on each individual were typically 31,000 s or frames of full 2-D and 3-D video representations of the pressure applied to the seat and back of the chair. Each of these files were converted from X-SENSOR files to ASCII files and then converted to Microsoft Excel files, in which a program to tally the relative counts of "person in chair" and "person applying pressure to back" was used. This summary data were then totaled across the shift with the means per hour being used in the software, SPSS (Version 15; SPSS, Chicago, IL), as a basis of statistical comparison. Counts were generated in Excel for each change in contact longer than 1 s, which exceeded a 50-mmHg peak pressure level.

The level was derived from our pilot study, which was conducted in the same manner and with the same equipment as described here for the full study. It was conducted to attempt to

establish a pressure threshold level for counting that would rule out simple side-to-side shifting while in normal contact with the seat or back. Simultaneous observation of pressure readings and participant behaviors was conducted for our first 5 participants independent of their shift of record used in the study. The pilot study also tested the first 5 participants for 5 days each rather than only 1 day to determine whether day-of-week effects were significant within 1 participant. According to one-way analysis of variance and the Bonferroni test, the day-of-week effect was not shown to be significant ($p > .21$), and the remaining test participants were screened only on one randomly selected day of the week each. This confirmed a similar pattern as seen by Fenety et al. (2000), who observed a consistent number of seat movements per day.

The general idea for the pressure count was to find the number of seat drops and back pulls needed to simulate chair use in the test lab. For current test methods, sand bags are dropped on the seat and a mechanical pull of the back is performed with the seat anchored. By placing the test chair from this study in the current test machine, we were able to compare peak pressure readings on our device with the current test forces to establish our pressure change of 50 mmHg/s. In addition to all full ingress-egress cycles (completely in and out of the chair for more than 1 s), we also found that our counts at this pressure level captured normal forward reaches (staying seated but leaning forward) and hard chair back returns (completely out and back in the chair), but minor back pressure (shifting around while maintaining contact) without release and reapplication (<50 mmHg/s) would not be counted. Additionally, significant weight shifts in the seat pan would be counted, but minimal movements or crossing the legs would not be counted.

The test chair was tethered to an anchored power cord that had minimal impact on mobility. To maintain maximum fidelity, given the cord and extra weight of the test chair, the caster test was performed at a later date with the participant's own chair, after replacing the participant's casters with the data-recording caster set.

Experimental Design and Analysis

The independent variable was group number (from BMI and weight). The following were the dependent variables: quantity of time per shift spent seated, quantity of cycles with the backrest, frequency of seat cycles per shift, and average distance rolled per shift. Statistical analysis of the means of each of the measures was compared to determine whether there is a significant difference between Group 1 and Group 2 workers, between males and females, and between Group 1 and Group 2 measures by gender, race, and job type. The null hypothesis that Group 1 means and Group 2 means for each measure were equal was established and tested with a normal independent-samples t test at $\alpha = .05$. Therefore, rejection of our null hypothesis with a standard Student t test for comparison of means would indicate a need to update test parameters on the basis of the impact of weight in our current population and the "how often" of the application of that weight.

RESULTS

Figure 3 displays the box plots for percent-age seated per 8-hr shift for the Group 1 versus Group 2. Table 4 shows that the difference in mean percentage seated per shift was significant ($p < .001$). The distribution shows a statistical outlier on the Group 2 distribution (Participant 10) as determined by the SPSS software. This participant was included in the analysis despite her own indication that her data collection day was not typical because of an above-average number of meetings. Additionally, conclusions were not affected by inclusion or exclusion of her data. The mean age for males was 38.8 years, and the mean age for females was 36 years.

In our sample, a statistical difference in age between groups ($p < .543$) and in age between genders was not seen ($p < .327$). Male and female means were compared for each study measure, and no significant differences existed. The only significantly different means ($p < .024$) by job type were clerical (mean of 74.4%) versus managerial (mean of 60.4%) in the percentage seated per shift. As seen in and from confidence intervals in Table 5, all means are significantly different at the .05 test level for

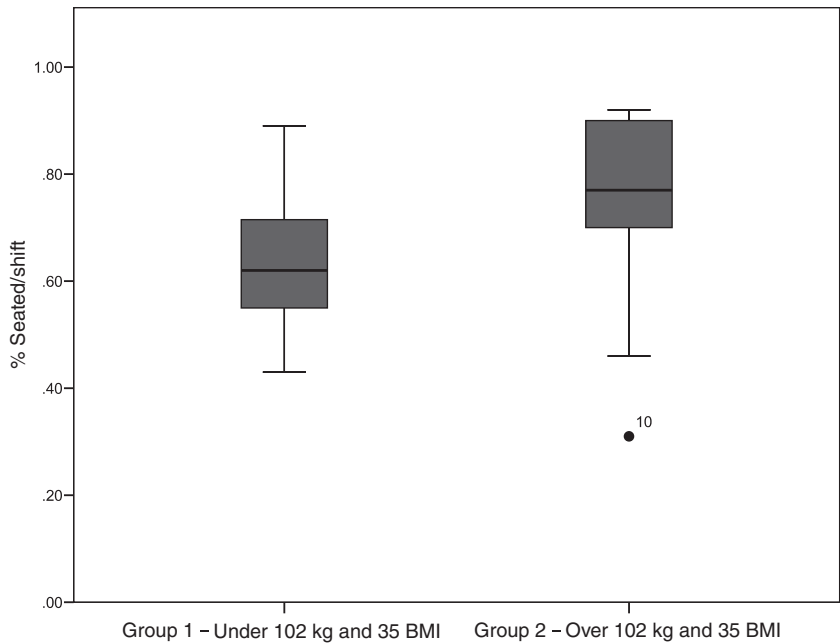


Figure 3. Distributions for percentage seated per shift by group (Participant 10, outlier per SPSS, was included in the analysis), $p < .001$.

TABLE 4: Test for Equality of Means for Group 1 Versus Group 2

Measure	<i>t</i>	<i>df</i>	<i>p</i> (two tailed)	Mean Difference	Standard Error of the Difference	95% Confidence Interval of the Difference
Roll distance (cm/hr), equal variances not assumed ^a	5.527	49	.000	322.5	58.3	[205, 439]
Seats per hour, equal variances assumed ^a	3.799	49	.000	2.58	0.68	[1.21, 3.95]
Backs per hour, equal variances assumed ^a	-3.275	49	.002	-18.33	5.61	[-29.66, -7.10]
Percentage seated per shift, equal variances assumed ^a	-3.476	49	.001	-0.135	0.038	[-0.213, -0.057]

^aLevene’s test for homogeneity of variance showed $p < .05$ in three of the four measures and $p < .563$ in roll distance per hour.

group versus seat cycles per hour, back cycles per hour, and percentage seated per shift.

Roll Distance Test

On separate days from the pressure distribution data collection, data were collected for roll distance of the chair casters to determine

whether BMI greater than 35 and or body weight in excess of the ANSI/BIFMA recommended 102.3-kg maximum would affect total distance rolled. Figure 4 shows that the mean roll distances for Group 1 (1,122 cm/hr) and Group 2 (800 cm/hr) are significantly different at the .05 test level with $p < .001$.

TABLE 5: Descriptive Statistics by Group

Measure	N	Mean	Standard Deviation	95% Confidence Interval for Mean	Minimum	Maximum
BMI (kg/m ²)						
Group 1	24	25.67	4.25	[23.87, 27.46]	17.47	33.65
Group 2	27	40.47	6.11	[38.05, 42.88]	35.13	54.68
Total	51	33.50	9.13	[30.93, 36.07]	17.47	54.68
Seats per hour						
Group 1	24	6.91	3.05	[5.62, 8.20]	2.13	11.65
Group 2	27	4.33	1.69	[3.66, 5.00]	1.97	8.71
Total	51	5.54	2.73	[4.77, 6.31]	1.97	11.65
Backs per hour						
Group 1	24	32.68	17.78	[25.17, 40.19]	6.40	70.36
Group 2	27	51.06	21.78	[42.44, 59.68]	6.60	81.41
Total	51	42.41	21.86	[36.26, 48.56]	6.40	81.41
Percentage seated per shift						
Group 1	24	0.63	0.127	[0.58, 0.68]	0.43	0.89
Group 2	27	0.77	0.148	[0.71, 0.82]	0.31	0.92
Total	51	0.71	0.15	[0.66, 0.75]	0.31	0.92
Roll distance (cm/hr)						
Group 1	24	1,123	188	[1,044, 1,202]	746	1,432
Group 2	27	800	224	[712, 889]	441	1,341
Total	51	952	262	[847, 1,057]	441	1,432

Note. BMI = body mass index.

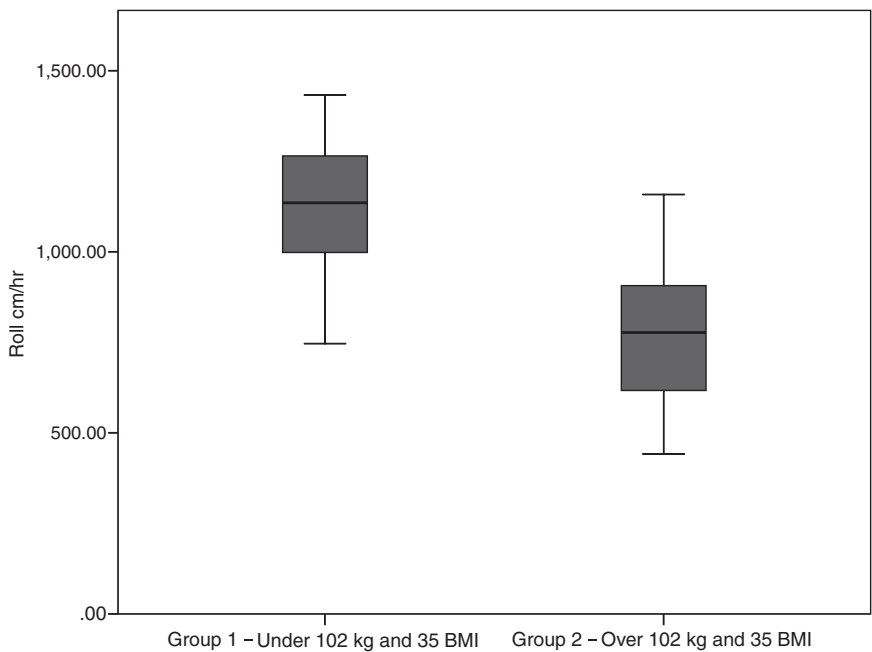


Figure 4. Roll distance distribution by group, $p < .001$.

DISCUSSION

During this study, it was determined that there were 54% more seat cycles per hour for Group 1 workers than for Group 2 workers. Specifically, a mean of 4.5 seat cycles per hour for Group 2 in and out of the chair and a mean of 6.9 seat cycles per hour for Group 1 in and out of the chair were recorded. There were 62% more back cycles per hour for Group 2 workers than for Group 1 workers. Specifically, a mean of 53 backrest cycles per hour for Group 2 and a mean of 32 backrest cycles per hour for Group 1 were recorded. There was 20% more seated time per shift for the Group 2 workers, who averaged 77% of the shift seated, compared with 64% of the shift seated for Group 1 workers. The practical impact of this finding on design and testing is significant, given that the heavier workers are spending the most time on average in their chairs. This means the largest static forces are being applied for the longest time. The current test does not simulate seated duration directly.

The ANSI/BIFMA X5.1-2002 test number of 120,000 cycles of 56.8-kg test drops into the seat to simulate 10 years of Group 1 use has been in place since the 1970s. During seat-use testing in this study, it was extrapolated from daily numbers that Group 2 seat cycles per decade would be 89,946, whereas Group 1 seat cycles per decade would be 138,348 per decade. The average number of seat cycles for all participants, extrapolated to the ANSI/BIFMA 10-year cycle, was 110,330. Given that the relative difference between our data and the current ANSI/BIFMA numbers is moderate, it is recommended that the testing cycles be increased to 140,000 to cover typical habits of all current workers.

Gender and age were not found to be a significant bias in this study. When analyzed by job type, the only parameter found to be significantly different was percentage seated per shift ($p < .024$). More specifically, management positions had less seated time than clerical and technical positions. This difference may not be significant to design, but it is significant in noting that the people most often called on to select and purchase seating for workers are also those who use their primary chair the least during a shift.

Caster Roll Test

The ANSI/BIFMA X5.1-2002 test standard calls for 100,000 cycles of 152 cm per cycle, or 76 cm down and 76 cm back per cycle, with a 102.2-kg weight centered over the five-leg base. This figure is 15,240,000 cm per 10 years in the test mode, or 152.4 km. The results from the caster field test found that even the average for both groups was more than 25,400,000 cm per decade, and for the peak- or highest-distance test participants for worst case, the distance was closer to 38,000,000 cm per 10 years, or 381 km. This figure is based on the average for 8 hr of test data per participant multiplied by 2,000 hr per year, times 10 years.

Roll distance was significantly different between Group 1 and Group 2 ($p < .001$). ANSI/BIFMA caster roll test criteria should be revised by increasing the weight load and the roll distance. Test load weight should increase to at least the latest 95th-percentile male numbers from the more recent, large-scale studies, such as CEASAR (Robinette et al., 2002). Roll distance should be at least three standard deviations from the group mean determined in this study. If a seating test for obese users is developed, it should have less roll distance than the Group 1 standard but a heavier weight. What is not provided for in the current ANSI/BIFMA test procedure is that the same base and casters that are roll tested are drop tested. Because of the uneven loading in real-time sitting, whereby one base leg is often loaded with most of the drop force from a person entering a seat, and then rolling, the roll and drop tests should be combined to more accurately approximate field conditions by drop testing after each roll cycle, with the same base and casters.

Limitations

The selection of a single representative chair may be a limitation of the interpretation of the results. We attempted to capture the current pressures on our test apparatus and then monitor for them in the field on the basis of current readings from test devices used according to the current standard. This method may or may not include data desired by designers for durability and comfort and would need to be further explored prior to sampling a larger group of chair

users. Our main purpose for using participant chairs for the roll test was to provide as much fidelity as possible. We would have preferred to have used the participants' own chairs in the other tests as well, but for safety and cost reasons, configuring 51 different chairs for that test would have been extremely expensive and complicated. Additionally, the sample size of 51 office workers would need to be significantly increased along with representation in each of the job types, geographic location, and potentially, time of year.

CONCLUSION

This research compared the office seating habits of Group 1 workers covered by current design and test standards, with a BMI of less than 35 and weighing less than 102 kg, and Group 2 workers, with a BMI greater than 35 and weighing more than 102 kg, to determine whether a difference in use patterns exists beyond the obvious force differences associated with heavier body weights. The purpose of the collection of these data was to create a basis for new standards used to validate design requirements of office seating in the areas of quality, durability, and performance. Group 1 was designed to be inclusive of workers covered by the current test document, whereas Group 2 was intended to go beyond that document.

In general, we found that Group 2 workers would enter and exit less, spend more time in their chair, and roll around more when compared with their Group 1 peers. To test seating designed for those weighing more than 102 kg and with a BMI greater than 35, we believe that the testing parameters should be more stringent in load time and frequency of back loads and load force and equal in roll distance to those in the current standards.

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KEY POINTS

- Workers weighing more than 102 kg and with a BMI greater than 35 spent 20% more time seated

per workday than coworkers who weighed less than 102 kg had a BMI less than 35.

- Workers weighing more than 102 kg with a BMI greater than 35 rolled their chair casters 23% less on average than workers below those levels.
- Current test standard assumptions for chair ingress and egress may no longer apply to information-age workers.

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